

1. a) AC voltage and current waveforms may be expressed mathematically as $v = V_{max}\sin\omega t$ or $i = I_{max}\sin\omega t$, while the RMS (root mean square) value of a sinusoidal voltage or current may be expressed as

$$V_{rms} = \frac{V_{max}}{\sqrt{2}}.$$

Therefore, the rms value of $v = 340\sin(628.318t + 0.5236)$ V is

$$V_{rms} = \frac{V_{max}}{\sqrt{2}} = \frac{340V}{\sqrt{2}} = 240.4163V$$

- b) Using the AC voltage/current formula given above, the angular frequency ω is identified as the coefficient of the independent variable t . ω is given in radians per second, and is connected to the frequency f in Hertz (cycles per second) by $\omega = 2\pi f$.

Therefore, the frequency of the given current is

$$\frac{\omega}{2\pi} = \frac{628.318}{2\pi} \approx 100\text{Hz}.$$

- c) The phase shift of a signal is incorporated into the basic formula for AC signals (using voltage as an example) as $v = V_{max}\sin(\omega t - \theta)$, where θ refers to the signal's phase shift in radians (positive phase angles shift the resultant waveform to the right along the time axis, while negative phase angles shift the resultant waveform to the left along the time axis).

For the given voltage, converting to degrees, $\theta_V = -30^\circ$, and for the given current, $\theta_I = -50^\circ$.

Therefore, the difference between the current and voltage phase angles is

$$\theta_{VI} = \theta_V - \theta_I = -30^\circ - (-50^\circ) = -20^\circ.$$

Note that since current leads voltage in this case, the load is capacitive, and the phase angle is negative, regardless of the sign produced by the given formula.

- d) The average value of a periodic signal, such as an AC voltage or current, may be found by averaging its value over a single period T of the signal. Averaging voltage over a single period $T = \frac{1}{f} = \frac{1}{100\text{Hz}} = 10\text{ms}$,

$$V_{ave} = \frac{1}{T} \int_0^T A\sin(\omega t - \theta_v)dt.$$

Since the sine and cosine functions are symmetric over each period, this integral formula will always evaluate to 0 regardless of the values of A , ω , or θ .

Therefore,

$$V_{ave} = 0V.$$

This is the reason rms values are more commonly used as a measure of the average value of an AC voltage or current.

- e) Load impedance Z is given by the sum of the real and imaginary components of impedance, resistance, R , and reactance, X ,

$$Z = R + jX,$$

and can be derived by expressing both voltage and current as phasors (assuming they have the same angular frequency ω and using the Ohm's Law relationship

$$Z = \frac{V}{I} = \frac{340\angle-30^\circ\text{V}}{100\angle-50^\circ\text{A}} = 3.4\angle-20^\circ\Omega.$$

Again, the phase angle is negative due to the capacitive load.

2. a) The power factor of a given load is given by the ratio between apparent and real power,

$$P.F. = \frac{P_{real}}{P_{apparent}},$$

where real and apparent power are given by

$$P_{apparent} = V_{rms}I_{rms} \quad (1)$$

$$P_{real} = V_{rms}I_{rms}\cos(\theta_{IV}). \quad (2)$$

To find the phase angle of the current, we may convert the complex impedance to phasor form,

$$Z = 4 - j2\Omega = \sqrt{20}\angle-26.5651^\circ$$

and use Ohm's Law in the form

$$I = \frac{V}{Z} = \frac{120\angle0^\circ\text{V}}{\sqrt{20}\angle-26.5651^\circ\Omega} = 26.8328\angle-26.5651^\circ\text{A}.$$

Thus, the phase angle is

$$\theta_{IV} = -26.5651^\circ.$$

Finally, finding apparent power

$$V_{rms}I_{rms} = 120\text{V} * 26.8328\text{A} = 3.22\text{kVA}$$

and real power

$$P_{real} = P_{apparent}\cos(\theta_{IV}) = 3.22\text{kVA} * 0.8944 = 2.88\text{kW}.$$

Therefore, the power factor of the given load is

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{2.88}{3.22} = 0.8944.$$

- b) Source current in phasor form was found in part (a). We cannot convert to the time domain since we are not given the angular frequency ω ; thus, the source current is

$$I = 26.8328 \angle -26.5651^\circ \text{ A}$$

- c) Real power was found in part (a),

$$P_{real} = 2.88 \text{ kW}$$

- d) Reactive power $P_{reactive}$ is given by

$$P_{reactive} = P_{apparent} \sin(\theta_{IV}) = 3.22 \text{ kVA} * 0.4472 = -1.44 \text{ kVAR}.$$

Once again, reactive power is given as negative due to the capacitive load.

3. a) Source frequency affects the reactance of both inductors and capacitors, defined as

$$X_L = 2\pi fL \quad (3)$$

$$X_C = \frac{1}{2\pi fC}. \quad (4)$$

Finding the reactance of the load inductor given,

$$X_L = 31.4034 \Omega.$$

Therefore, complex impedance in phasor form is given as

$$Z = 3.5 + j31.4034 \Omega = 31.5978 \angle 83.6405^\circ \Omega$$

and load current is found as

$$I = \frac{V}{Z} = \frac{120 \angle 30^\circ \text{ V}}{31.5978 \angle 83.6405^\circ \Omega} = 3.7977 \angle 53.6405^\circ \text{ A}$$

- b) Collecting needed values,

$$P_{apparent} = V_{\text{rms}} I_{\text{rms}} = 84.8582 \text{ V} * 2.6854 \text{ A} = 227.8765 \text{ kVA}$$

$$\theta_{IV} = \theta_I - \theta_V = -53.6405^\circ - 30^\circ = -83.6405^\circ$$

$$P_{real} = P_{apparent} \cos(\theta_{IV}) = 227.8765 \text{ kVA} * 0.1108 = 25.2411 \text{ kW},$$

and the load power factor is

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{25.2411}{227.8765} = 0.1108.$$

- c) Real power was found in part (b),

$$P_{real} = 25.2411 \text{ kW}.$$

d) Reactive power is

$$P_{reactive} = P_{apparent} \sin(\theta_{IV}) = 226.4742 \text{VAR}.$$

4. a) In this case, with a source frequency of 50Hz, the reactance of the inductor will change to be

$$X_L = 26.1695 \Omega.$$

Recalculating the circuit's complex impedance and finding the load current,

$$I = \frac{V}{Z} = \frac{120 \angle 30^\circ \text{V}}{26.4025 \angle 82.3823^\circ \Omega} = 4.5450 \angle -52.3823^\circ \text{A}.$$

- b) The load power factor is then found by recalculating the real and apparent power and finding the ratio between them,

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{36.15 \text{W}}{272.7 \text{VA}} = 0.1326$$

- c) Real power is given in part (b) as

$$P_{real} = 36.15 \text{W}.$$

d) Reactive power is

$$P_{reactive} = P_{apparent} \sin(\theta_{IV}) = 270.2933 \text{VAR}.$$

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